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To cite this article: Toshihiro Yokoo & David Levinson (2019) Measures of speeding from a GPS-based travel behavior survey, Traffic Injury Prevention, 20:2, 158-163, DOI: [10.1080/15389588.2018.1543873](https://doi.org/10.1080/15389588.2018.1543873)

To link to this article: <https://doi.org/10.1080/15389588.2018.1543873>



Published online: 19 Mar 2019.



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Measures of speeding from a GPS-based travel behavior survey

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ABSTRACT

Objective: Lacking information about actual driving speed on most roads in the Minneapolis–St. Paul region, we determine car speeds using observations from a Global Positioning System (GPS)-based travel survey. Speed of travel determines the likelihood and consequences of collisions. We identify the road segments where speeding occurs. This article then analyzes the relationship between link length, traveler characteristics, and speeding using GPS data collected from 152 individuals over a 7-day period as part of the Minneapolis–St. Paul Travel Behavior Inventory.

Methods: To investigate the relationship, we employed an algorithm and process to accurately match the GPS data with geographic information system (GIS) databases. Comparing actual travel speed from GPS data with posted speed limits, we measure where and when speeding occurs and by whom. We posit that link length and demographics shape the decision to speed.

Results: Speeding is widespread under both high speed limits (e.g., 60 mph [97 km/h]) and low speed limits (less than 25 mph [40 km/h]); in contrast, speeding is less common at 30–35 mph (48–56 km/h). The results suggest that driving patterns depend on the road type. We also find that when there are many intersections, the average link speed (and speeding) drops. Long links are conducive to speeding. Younger drivers and more educated drivers also speed more, and speeding occurs more often in the evening.

Conclusions: Road design and link length (or its converse, frequency of intersections) affect the likelihood of speeding. Use of increasingly available GPS data allows more systematic empirical analysis of designs and topologies that are conducive to road safety.

ARTICLE HISTORY

Received 20 May 2018
Accepted 29 October 2018

KEYWORDS

GPS; speeding; speed limit; link length; travel behavior

Introduction

Speed of travel determines the likelihood and consequences of collisions. Yet little is known about the extent and magnitude of the speeding problem on most roads due to a lack of systematic data. Speeding behavior, driving faster than the speed limit, is affected by the road environment, the vehicle, and the driver (Kanellaidis 1995; Sila-Nowicka 2018). Although other researchers (Debnath et al. 2014; Kanellaidis et al. 1995; Lai and Carsten 2012; Saifizul et al. 2011) have analyzed reasons for speeding considering driver behaviors, traffic conditions, and vehicles, we posit that the road network structure in the form of link length also has an effect on speeding.

Our article corroborates results showing that speeding occurs more often at the upper and lower levels of speed limit ranges (Lai and Carsten 2012), which suggests that drivers change their speed limit compliance patterns depending on the road type and that speed limits that are not consistent with driver expectations of appropriate speed are more likely to be disregarded. When there are many intersections on the road network, the average speed on the road linking those intersections is reduced due to traffic signals, traffic conditions, and vehicles approaching from other directions.

In order to analyze the relationship between link length and speed limits and speeding, we need to collect a large amount of

traffic data over a broad area. Historically, traffic detectors and area studies have been used to analyze traffic data. However, the amount of data on local roads from such studies is small and infrequent. Therefore, in this article, we take advantage of Global Positioning System (GPS) data, which contain time and position (latitude, longitude, altitude) and allow us to compute travel speed. Though this set of data comes from a 2010–2011 travel survey, as GPS data become more widespread with the diffusion of smartphones and other devices, we anticipate that this kind of analysis will become increasingly common.

In this article, we analyze the relationship between demographics, link length, and speed limit and the degree of speeding from a GPS data set. To investigate the spatial relationships, we employ an algorithm to accurately match the GPS data and geographic information system (GIS) maps. We then regress GPS speed observations with speed limits, link length, and demographic data.

Methods

Hypotheses

We posit that speeding is affected by environmental and personal characteristics. We enumerate specific hypotheses below:

Hypothesis 1. H1: Link length—Long link length (long spacing between intersections) is correlated with speeding (Baruya 1998) because there are fewer conflicts and less requirements for deceleration.

Hypothesis 2. H2: Hierarchy (road type)—Position near the top of the hierarchy of roads (e.g., freeways and limited-access facilities) is correlated with speeding (Stradling et al. 2002). Compared to local roads, the driver is less affected by external factors on freeways. This is operationalized by looking at speed limit category. We note that there are locations that curve, and gradients and other factors may also affect speed limits, though this is not particularly common in the Minneapolis–St. Paul region. Other measures of hierarchy, such as formal classification, might also be used.

Hypothesis 3. H3: Age—Younger drivers speed more regularly (Cestac et al. 2011).

Hypothesis 4. H4: Gender—Male drivers speed more regularly (Laapotti and Keskinen 2004).

Hypothesis 5. H5: Education level—Better educated drivers will speed more. Education is generally correlated with income, which often shows this effect (Shinar et al. 2001).

Hypothesis 6. H6: Time of day—Speeding occurs most often in periods unaffected by traffic congestion and when enforcement is lower. This includes the evening and overnight periods (Greaves and Ellison 2011).

Degree of speeding

We calculate speeding behavior from 2 perspectives:

- Whether speeding occurs or not (percentage of speeding)—When the driving speed is over the speed limit, the value is 1; otherwise, the value is 0.
- How much the driver exceeds the speed limit (degree of speeding)—The percentage of speeding is critical because driving at 60 mph in a 40 mph speed zone and driving at 41 mph in a 40 mph speed zone are identical outcomes. Therefore, we also analyze another speeding behavior by calculating how much each GPS data point exceeds the speed limit. Degree of speeding is defined in Equation (1).

$$\text{Degree of Speeding} = \frac{\text{Driving Speed} - \text{Speed Limit}}{\text{Speed Limit}}. \quad (1)$$

When the value is 1, driving speed equals the speed limit, and when the value is greater than 1, the data show speeding.

Data

This research uses GPS data from the 2010 Twin Cities (Minneapolis–St. Paul area) Travel Behavior Inventory (TBI2010) administered by the Metropolitan Council between 2010 and 2012 (most data were collected in 2011) that has been used in a variety of studies (Charleux 2018; Ehrlich 2013; Ermagun and Levinson 2016; Guthrie and Fan 2016; Metropolitan Council of the Twin Cities 2015;

Schoner and Lindsey 2015; Schoner et al. 2016; Wang and Levinson 2016). The TBI is a large, roughly decennial, travel survey of a 1% sample of households in the Twin Cities region. The survey collects data on household and individual socioeconomic and demographic information and collects travel diary information (start and end time, purpose, mode, accompaniment, vehicle) about each trip made by each member of the household on a single day. To supplement the data and prepare for future data collection, a small sample of participants in the TBI was also equipped with a GPS device (Tang and Levinson 2018). Each subject in the GPS component of the TBI carried a GPS pendant for 7 days. The raw GPS data contain the following trip information: Speed (km/h); longitude, latitude, altitude (m); date (year/month/day); time (h/min/sec); distance (m); course (°); number of satellites; and horizontal dilution of precision. Seven consecutive days of movement with 1-s recording intervals were collected.

Because the initial GPS data did not record each trip and the time that the GPS was disconnected, we define the trip based on the time difference.

To process the data, we undertake the following steps (Yokoo 2016):

1. Divide the initial GPS data into trips when a time difference between GPS data locations exceeds 300 s.
2. Remove trips with either end outside Minneapolis–St. Paul region (7 counties).
3. Classify mode.
4. Remove speeds less than 5 km/h.
5. Convert coordinate systems.
6. Match trip ID with survey data.
7. Match GPS points with GIS map. The map-matching algorithm uses a series of GIS procedures, buffers, and exclusions to ensure that every observation is correctly matched to a link and avoid false positives (at the expense of excluding points that could be matched; false negatives), detailed in Yokoo (2016).

The GPS data alone do not contain personal information (e.g., gender, age); therefore, we complement the GPS data with the associated records from the TBI2010 Household Interview Survey (e.g., household data, personal information, trip data). These data can be matched on person ID. Among 274 GPS subjects (drawn from 250 households), 152 travelers have IDs matching the survey and no other missing data, satisfying conditions that both ends of the trips are within the study region, speeds were greater than 5 km/h, the traveler is over 16, and mode was an automobile, allowing us to analyze these participants. We append the GPS data with personal information describing each subject's gender, age, and education. The other travelers were excluded from further analysis.

To understand the actual speed limit data, we use a GIS database maintained by the Metropolitan Council and NCompass (NCompass Technologies 2013) that covers the majority of road segments in the Twin Cities' 7 county metropolitan area. The database contains 290,231 links, and

each one-way link has several attributes (e.g., speed limit, link length, street name).

To conduct the analysis, we use QGIS Ver. 2 (Sutton and Dassau 2015), an open-source geographic information system. The GPS data are matched to the GIS database using the procedure outlined in Yokoo (2016). In brief, we aligned GPS points with links on the GIS database. Where there were clear matches and no ambiguities, the matches were confirmed. Where GPS points were outside a buffer of the links, they were excluded. Where GPS points could have been associated with more than one link (fell in the buffer of 2 or more links), they were excluded. As a consequence, 51.0% of GPS data were removed in our map-matching algorithm in order to ensure no false positives. We combine multiple GPS points from the same driver on the same trip on the same link. GPS points were recorded every second; however, the speeding patterns of drivers on the same link are similar from second to second. Therefore, we integrate multiple GPS points for each link for each driver for each trip into a single analysis record; we separate the link data depending for each driver and for each trip because driving environments and driving patterns are different based on other trip conditions and for different drivers. We thus analyzed 1,172,103 GPS points, which were aggregated to 115,536 link-trip records from 152 drivers across 24,262 links, making 2,560 trips in total. Future research will aim to reduce the amount of data excluded (false negatives) using more accurate map-matching methods (Lou et al. 2009).

Though the advantages of GPS data are clear (large diverse sample of links over a broad area with varying conditions and many drivers), it has disadvantages. GPS does not present a complete census of drivers on a particular link over a given study period the way more traditional point-based speeding measurements might. Therefore, there is the possibility of bias emerging, especially if the sample favors those who voluntarily use GPS, which might include more drivers who are male, are young, have a high income, and are technologically inclined (and perhaps more likely to speed as a result). As GPS samples become larger, this issue is likely to be mitigated.

The data were analyzed using the statistical package R Ver. 3.0. In order to eliminate the fixed effect of individuals prior to estimation, we use a dummy variable regression wherein we introduce a dummy variable for each individual along with the independent variables.

Results

Descriptive analysis

We regress the relationship between link length, speed limit, demographics, and speeding. As Figure 1 illustrates, most of the data are concentrated under 30, 35, 40, and 55 mph speed limits. Moreover, relatively few points are observed in locations with low (less than 20 mph) and high (70 mph) speed limits. (Note that some travel that is retained in the analysis occurred on private roads or in parking lots, which explains travel on roads with 5 mph speed limits.) The difference between GPS points and link samples shows 2 different types of bias. The first has to do with time; that is, GPS

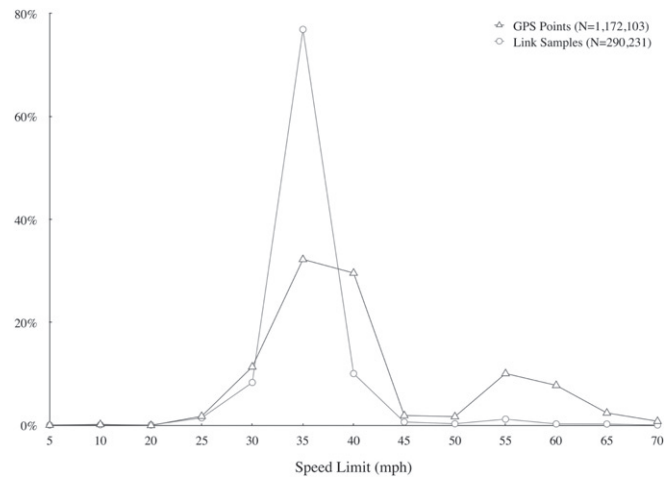


Figure 1. Distribution of speed limits.

points will sample links more heavily with low speed limits, because vehicles are traveling more slowly, so the same frequency will collect more points on those links. The second has to do with link length. If links with higher speed limits tend to be longer, those links have more samples per link, after controlling for speed. Clearly the first point is dominant, because link samples are far more concentrated at 35 mph than GPS points are.

Figure 2 shows the percentage of speeding for each speed limit. Overall, 23.3% of GPS observations exceed the speed limit. It can also be seen that speeding behavior is significant at low speed limits (e.g., from 5 to 25 mph) and high speed limits (e.g., from 55 to 65 mph). Most participants encountered speed limits between 25 and 65 mph while driving.

The 20 mph speed limit stands at 88.8%, and it is followed by 5 and 25 mph speed limits at 80.9 and 73.3%, respectively. However, the amount of data in low speed limits is relatively small; therefore, more data would improve the reliability of this result. In addition to low speed limits, speeding in high speed limit ranges (from 45 to 70 mph) is also higher than average. Speed limits from 30 to 40 mph have less speeding than lower and higher speed limits, and the 30 mph speed limit is the lowest, with only 10.1% speeding.

We analyze speeding by time of day in Figure 3. Overall, speeding behavior varies by time of day and the percentage of speeding at night exceeds that during the day, with a peak at 4 a.m. (64.7%), followed by 5 a.m. and 11 p.m. (40.1 and 39.8%, respectively). On the other hand, there is less speeding at 8 a.m. and 9 a.m., which is likely related to the effect of rush hour. It is assumed that drivers cannot drive faster because of traffic conditions. Although the percentage of speeding at 3 a.m. is considerably lower than that other times (3.7%), the number of participants traveling at this time is low ($N=3$) and it appears that this result might be significantly influenced by individual factors.

Clearly, driving styles vary individually; some drivers are chronic speeders (even while knowingly being monitored by a GPS unit, though it was clearly not aimed at speed enforcement), whereas others consistently travel below the speed limit, as shown in Figure 4. However only a handful of subjects avoided speeding for the entire 7-day period.

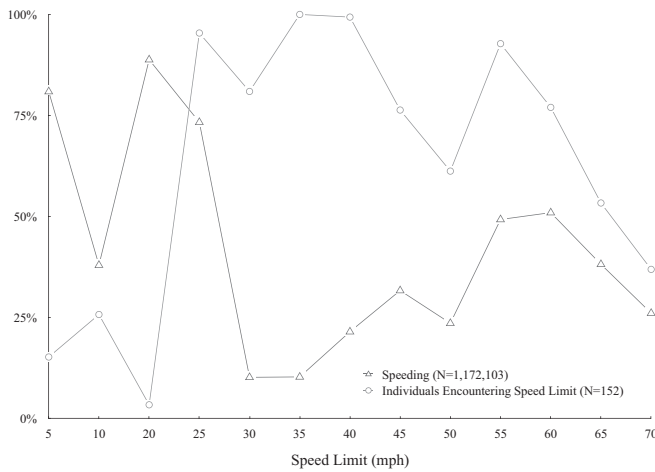


Figure 2. Distribution of speeding across speed limits.

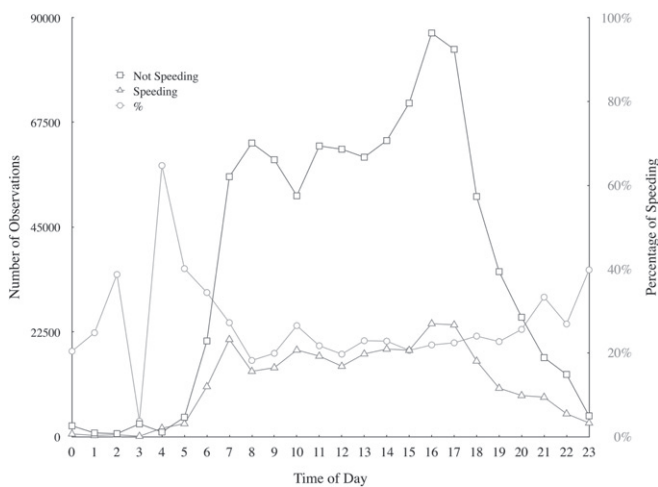


Figure 3. Distribution of travel and speeding by time of day.

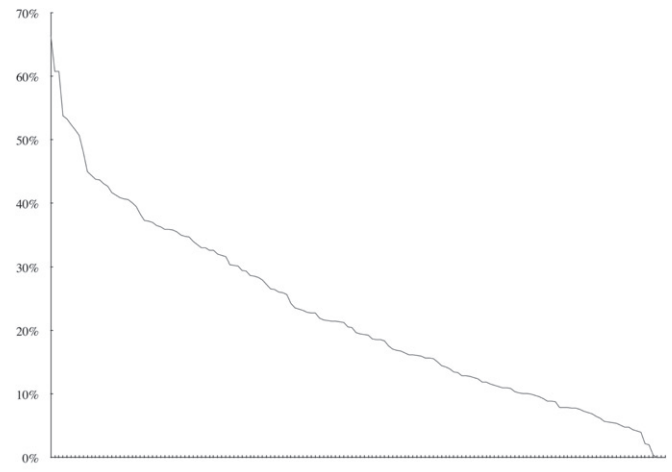
Statistical results

Next, we analyze the relationship between several variables (link length, speed limit, and personal information) and speeding using an ordinary least squares regression with individual fixed effects. Table 1 shows the results of the regression analysis. We do not report individual driver statistical estimates here, but these estimates can be found in Yokoo (2016).

Though the t statistics for 40, 45, and 50 mph speed limits; age 55–64; and male are small, the t statistics for other variables are large. Overall, F-tests shows that the F value is 280.5 and the P value is approximately zero. Therefore, this model shows high significance. Overall the R^2 is 0.29. A positive coefficient indicates that when the street length is long, speeding is likely to occur, corroborating theory and the descriptive statistics.

Degree of speeding varies by speed limit. It occurs most at 25 mph and on links with 55, 60, and 65 mph limits. However, 30 and 35 mph speed limits are negatively associated with speeding.

Persons who are young, drive overnight (12:00 a.m. to 6:00 a.m.), and/or are educated speed more, corroborating our hypotheses. Those in the age category 25–34 speed most frequently. In contrast with earlier results, we find no



Individuals, Sorted by the Amount of Speeding

Figure 4. Individual travelers sorted by frequency of speeding.

evidence that males speed more often; therefore, the variable is not statistically significant.

A correlation test finds that the independent variables are not problematically correlated. Speed limit and street length have a correlation (r) of 0.51, so speed limit was introduced as a series of discrete dummy variable, for which the correlation with street length ranges from $r=0.21$ at a speed limit of 25 mph to $r=0.30$ at 60 mph. The demographic variables are independent of each other.

Discussion

This article investigates GPS-based speed data from 152 subjects to examine the relationship between link length, speed limit, and speeding. The most pertinent findings from the results are that speeding behavior is significant at both low and high speed limits, and long link length is correlated with speeding. Regression analysis finds that persons who are older, drive in the afternoon, and/or are less educated speed less than those in younger age groups, at other times of the day, and/or those who are more educated.

Other studies mention that elderly, female, and uneducated drivers who do not often drive on interurban roads are likely to follow the speed limit (Kanellaidis et al. 1995; Lai and Carsten, 2012). We did not see differentiation by gender, in contrast with our own previous work (Chi et al. 2013). To corroborate our findings, this result needs to be replicated in further studies.

Speeding behavior across speed limits has been previously analyzed (Lai and Carsten 2012); according to our previous study results as well as the results of this study, speeding illustrates a V shape across all speed limits. In a previous study, speeding occurred less often at a 60 mph speed limit (Lai and Carsten 2012), whereas our result shows less speeding at a 30 mph speed limit. We posit that the difference in road characteristics might affect the result.

Future research should use larger sample sizes with observations from more GPS units. We expect that this will become easier over time as GPS units and the associated data become ubiquitous.

Table 1. Degree of speeding, fixed effects model.^a

Independent variable	Estimate	t Statistic	Significance
(Intercept)	8.39E-01	35.082	**
Street length (m)	1.28E-04	34.736	**
Posted speed limit (mph)			
25	6.59E-01	53.206	**
30	−3.70E-02	−3.125	*
35	−1.18E-01	−10.162	**
40	−8.41E-03	−0.727	
45	−3.29E-03	−0.255	
50	8.43E-03	0.636	
55	1.06E-01	9.164	**
60	9.92E-02	8.538	**
65	3.51E-02	2.823	*
Age			
25–34	7.49E-02	4.963	**
35–44	−1.06E-01	−3.933	**
45–54	−2.18E-01	−8.49	**
55–64	−3.58E-02	−1.513	
65–74	−1.57E-01	−7.345	**
75–84	−5.70E-02	−3.927	**
85+	−3.85E-02	−1.474	
Gender			
Male	−1.03E-02	−0.58	
Education			
Some college or associates degree	1.52E-01	9.02	**
Bachelor's degree or higher	1.35E-01	4.676	**
Time of day			
6:00 a.m. to 12:00 p.m.	−3.23E-02	−4.726	**
12:00 p.m. to 6:00 p.m.	3.57E-02	−5.254	**
6:00 p.m. to 12:00 a.m.	−2.30E-02	−3.285	*

^aMultiple $R^2 = 0.2851$; F-statistic = 280.5 $P < 0.000$. Individual fixed effects not reported in the table. The following levels were excluded to avoid the dummy variable trap: Posted speed limit: 70 mph; age: 18–24; education level: high school graduate and vocational/technical training; time of day: overnight (12:00 a.m.–6:00 a.m.).

* $P < .01$. ** $P < .001$.

Future analysis should address other measures of roadway geometry and network structure, including many topological and network scaling measures that are now more readily obtained through standardized scripts (Boeing 2017). Two specific hypotheses relate to discontinuity of speed limits (we anticipate that a large discontinuity would be correlated with speeding) and curvature (we expect that straight roads are correlated with speeding).

The results might be useful to decision makers, engineers, and planners. Speeding problems are widespread in many areas, and speeding increases crash risk. Being able to identify the types of roads that lead to speeding should help road designers rethink their designs in a way that would mitigate the issue.

It might be useful to infer that because long links are associated with speeding, new street designs should have shorter lengths, and links should be shortened with breaks like intersections or traffic calming measures. Because these data are cross-sectional rather than longitudinal, they cannot confirm that such changes would reduce speeding, though the results are supportive. These findings could be corroborated if followed up by carefully designed studies showing before and after effects. New urbanist street designs, including denser street networks (the antithesis of long lengths), have been shown to reduce crashes, which is posited to be due to lowered speeds (Marshall and Garrick 2011).

Adjusting speed limits may also reduce speeding but may or may not increase safety depending on the direction of adjustment. Lowered speed limits on non-limited-access roads along with improved designs to encourage lower

speeds and better (including automated) enforcement, or even in-vehicle speed limiters, may help in this regard (World Health Organization 2015).

A higher speeding rate among young adults was once again found in this study, which may be ameliorated with better enforcement and education and more restrictive licensure (Chapman et al. 2014).

Data availability

The GPS data sets analyzed in the current study are available from the Transportation Secure Data Center repository, <https://www.nrel.gov/transportation/secure-transportation-data/tsdc-minnesota-travel-behavior-inventory.html> (Metropolitan Council of the Twin Cities 2017). The street maps are available from MetroGIS (NCompass Technologies 2013).

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